Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment

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Abstract

We propose a unified multimodal architecture for mental health assessment that jointly learns from depression, sentiment, emotion, and personality across three public benchmarks: DAIC-WOZ (clinical interviews, $n = 189$), CMU-MOSEI (YouTube videos, $n = 22{,}777$), and ChaLearn First Impressions (video clips, $n = 10{,}000$). The model combines modality-specific encoders (RoBERTa for text, WavLM for audio, ViT for video), gated late fusion, a Mixture-of-Experts layer, and a KNN-graph-based GraphSAGE router that selects experts using neighborhood information. Graph-based routing does *not* improve over a non-graph MMoEEx baseline in this setting: across five leakage-safe graph configurations, DAIC depression AUROC (0.489–0.551), MOSEI sentiment CCC (0.510–0.517), and MOSEI emotion AUROC (0.712–0.734) are statistically indistinguishable from the non-graph baseline (0.493, 0.493, 0.734 respectively), and FI personality CCC is consistently *worse* with graph routing (0.540–0.553 vs. 0.578 without). A graph-homophily diagnostic shows the routing graph carries no more DAIC-label signal than random pairing, while it carries substantial signal for MOSEI and FI that the router nonetheless fails to convert into a performance gain; a learned per-task routing weight and a $K \in \{5, 10, 15, 20\}$ sensitivity sweep both fail to recover a benefit. We report this as a rigorously investigated negative result. Separately, cross-attention fusion does not reproduce a reported gain over gated fusion under our evaluation protocol, and LLM-based encoders (Mistral-7B, CLAP, LLaVA) substantially outperform both classical encoders and graph routing on every task. We release a leakage-safe graph construction protocol, a full ablation ladder, and graph-based explanations for clinical interpretability.

1 Introduction

Mental health assessment from multimodal signals is a cornerstone challenge in affective computing and clinical informatics. Depression, the primary focus of this work, affects over 280 million people worldwide [1] and is routinely assessed through clinical interviews that combine speech, facial expressions, and verbal content. Meanwhile, related affective signals—sentiment, emotion, and apparent personality—provide complementary supervision signals that can enrich mental health models through multitask learning [2, 3].

Our central hypothesis has two parts, which we evaluate separately rather than conflating. First, we hypothesize that mental health assessment can be made more *transparent* by characterizing a person's affective and psychological profile sentiment, emotion, and apparent personality jointly with a depression assessment, rather than reporting a single opaque depression score in isolation: the auxiliary constructs give a clinician additional, more legible context (e.g., is a low mood reading accompanied by high negative sentiment and low extraversion, or is it isolated?) for interpreting the primary prediction. Second, we hypothesize that a graph over similar individuals can provide both a *visual* (subgraph diagrams, per-modality attribution plots) and a *machine-readable* (structured case records: SHAP values, neighbor identities and distances, routing weights) account of which factors and which comparable cases informed a given prediction, complementing the auxiliary-construct characterization with instance-level explanation. This is a claim about transparency, evaluated separately from the further hypothesis that graph-based *routing* improves raw predictive performance (Section 6) a model can be well-explained without being more accurate, and our results show these are empirically distinct questions in this setting: the performance hypothesis is not supported, while the transparency mechanisms (multi-construct characterization, graph-based visual and machine-readable explanation) are implemented and evaluated for faithfulness in Section 8. We do not claim that apparent personality and clinical depression measure the same construct personality is auxiliary context, not a depression proxy. The proposed system is intended as a decision-support tool and should not replace clinical diagnosis; all results require prospective validation before clinical deployment.

Figure 1 gives an overview of the full experimental pipeline, from data and feature extraction through the baseline ladder, the graph-routing investigation (Section 6), and the downstream LLM, calibration, domain-adaptation, and explainability analyses.

This work makes three contributions:

1. **A rigorously investigated negative result for graph-gated MoE routing**: across five leakage-safe graph configurations, a K-sensitivity sweep, and a learned-routing-weight variant, graph-based expert routing does not improve over a non-graph MMoEEx baseline, and degrades FI personality prediction specifically. We support this with a graph-homophily diagnostic that localizes *why* (Section 6), rather than reporting a single negative number without investigation.
2. **Leakage-safe graph construction protocol**: Three graph construction modes (inductive, split-local, transductive) under strict subject-independent splits, with formal validation that prevents test-label leakage through graph edges.
3. **Comprehensive empirical analysis**: A systematic ablation ladder spanning unimodal baselines, three fusion strategies, MoE variants, five graph configurations, and LLM encoder replacements, including a comparison showing LLM-based encoders outperform both classical encoders and graph routing on every task.

2 Related Work

Multimodal Fusion. Early fusion for mental health used simple concatenation [2]. Later approaches adopted late fusion with learned modality gates [4] or Low-Rank Multimodal Fusion (LMF) [5]. Cross-modal attention mechanisms have been proposed for depression detection [6], but in our experimental configuration, we do not observe the reported +0.041 AUROC gain.

Mixture of Experts. The sparsely-gated MoE layer [7] routes each input to a subset of experts. Multitask MoE (MMoE) extends this with task-specific gates [8], and MMoEEx adds expert exclusivity and orthogonality regularization [9]. We build on MMoEEx but add graph-based routing.

Graph Neural Networks for Routing. GraphSAGE [10] performs inductive representation learning on graphs. We use GraphSAGE as a router: aggregating features from K nearest neighbors to produce a graph context vector that modulates expert weights. Wang et al. [11] similarly combine GNN-based experts with explicit diversity modeling, though as GNN experts rather than a graph-conditioned router over standard MLP experts as here. Whether a KNN routing graph is useful depends on its label homophily/informativeness [12]; when it is not, graph rewiring or learned graph structure [13] are natural remedies we discuss in Section 9. Task-affinity-aware grouping [14] and expert-choice

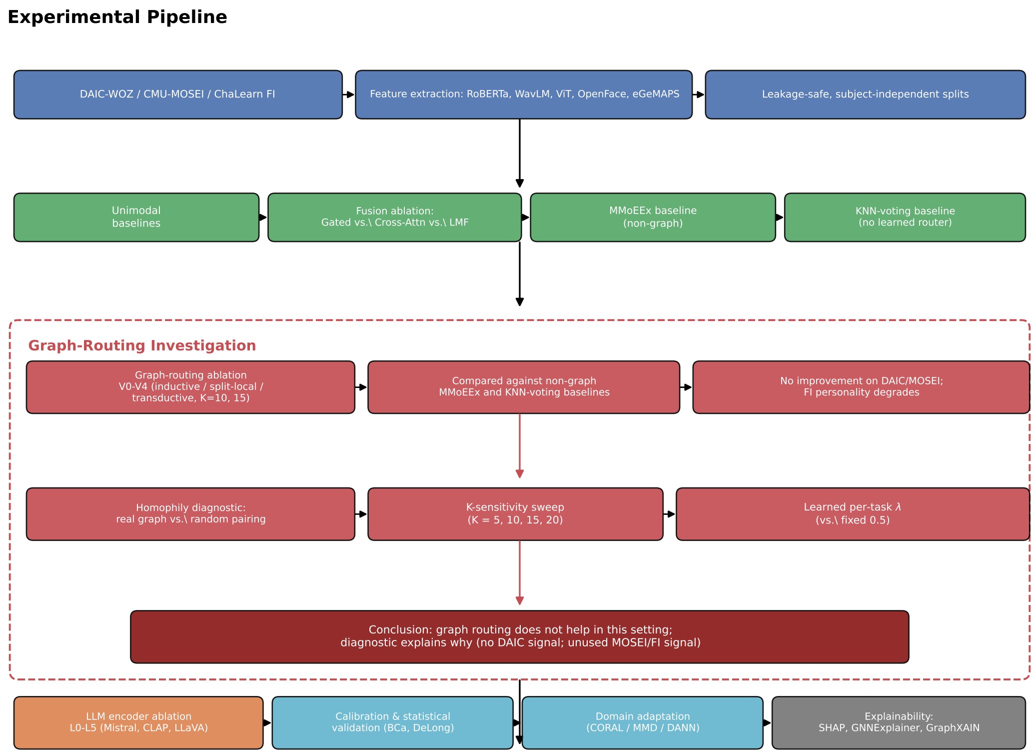

Figure 1: Experimental pipeline overview. The graph-routing investigation (dashed box) is the central empirical question of this paper: five ablation variants are compared against non-graph baselines, and a homophily diagnostic together with two targeted fixes (K-sweep, learned λ) explains the resulting negative finding.

routing [15] are two further candidate fixes we did not yet implement.

LLM-Enhanced Encoders. Large language models have been applied to depression detection via instruction-tuned text encoders [16] and audio-language models [17]. We evaluate LLM-enhanced encoders in our ablation track (levels L1–L5).

Multi-Construct Characterization and Graph-Based Interpretability. Interpretability for mental health assessment has been pursued by jointly modeling depression alongside related affective constructs, and by grounding graph structure in psychologically meaningful units: Vyalla et al. [18] explicitly distinguish personality trait context from acute depression symptoms via a psychologically-grounded, interpretable graph model, closely related to our multi-task characterization hypothesis (Section 1). For explaining LLM-based reasoning specifically, chain-of-thought and related structured-reasoning techniques have been shown to improve mental-health text classification [19], but can also be unfaithful or reduce reliability when reasoning is inserted into the clinical prediction pipeline itself [20] a distinction we return to in Section 8, where LLM narratives are used purely as *post-hoc* explanation of an independently-computed prediction, not as part of the prediction process.

Depression Detection on DAIC-WOZ. The DAIC-WOZ corpus [21] contains 189 clinical interviews with PHQ-8 labels. State-of-the-art methods include: Burdisso et al. GCN (F1=0.85) [22]; Zhang et al. MIL (AUROC=0.78) [23]; Niu et al. multimodal fusion (F1=0.92) [24]; Dai et al. full fusion (F1=0.96) [25]. Direct metric comparison is limited: most papers report F1 while we report AUROC.

3 Datasets

DAIC-WOZ contains 189 sessions (107 train, 35 val, 47 test) of human-agent interviews for depression assessment. Each session has a PHQ-8 binary label (depression ≥ 10). We use subject-independent splits: no participant appears in more than one split. Features: RoBERTa-base text (768-dim), WavLM-large audio (1536-dim), eGeMAPS (88-dim), ViT-B/16 video (1536-dim), OpenFace AUs (70-dim).

CMU-MOSEI [26] contains 22,777 utterance-level samples (16,265 train, 1,869 val, 4,643 test). Labels: continuous sentiment and multi-label emotion (happiness, sadness, anger, fear, disgust, surprise). MOSEI is 120× larger than DAIC, mitigated by temperature-balanced sampling (T=3.0).

ChaLearn First Impressions [27] contains 10,000 video clips (6,000 train, 2,000 val, 2,000 test) with Big-Five apparent personality labels. Personality is not the same construct as clinical depression; FI serves as auxiliary supervision.

Leakage Protocol. All splits are subject-independent. The graph is constructed separately for each split (split-local mode) or only on training data (inductive mode) to prevent test-label leakage through graph edges.

Table 1: Notation.

Symbol	Meaning
K	Nearest neighbors (graph)
M	Number of experts (8)
d	Embedding dimension (256)
r	Low-rank bottleneck (8–16)
λ	Graph routing weight
σ_t	Task-specific uncertainty
G	KNN graph
r_i	Graph routing vector

4 Architecture

Figure 2 gives an overview of the full pipeline.

Modality Encoders. Each modality $m \in \{\text{text}, \text{audio}, \text{video}\}$ is encoded by a pretrained encoder f_m, producing embedding h_i^m. A projection layer P_m maps to a common $d = 256$-dimensional space:

$$h_i^{m,\text{proj}} = P_m(f_m(x_i^m)) \in \mathbb{R}^{256} \tag{1}$$

Gated Late Fusion. Given projected modality embeddings, we apply a modality mask $m_i \in \{0,1\}^3$ and compute learned gate weights:

$$\alpha_i = \text{softmax}(W_\alpha \cdot \text{concat}(h_i^{\text{text}}, h_i^{\text{audio}}, h_i^{\text{video}}) + b_\alpha) \tag{2}$$

$$h_i^{\text{fused}} = \sum_m \frac{\alpha_i \odot m_i}{\|\alpha_i \odot m_i\|_1 + \epsilon}[m] \cdot h_i^{m,\text{proj}} \tag{3}$$

Each dataset is evaluated under a fixed, dataset-native routing policy chosen from our fusion ablation (Section 6): DAIC uses text-only routing ($h_i^{\text{fused}} = h_i^{\text{text}}$), FI uses video-only routing, and MOSEI uses the full gated fusion above. This is a deliberate architectural choice, not a missing-data fallback: text alone is DAIC's strongest single modality and video alone is FI's, so routing each dataset through its best-performing modality avoids diluting the fused representation with weaker signal on datasets too small to learn a reliable gate over all three. A direct consequence, verified in Section 8, is that a modality outside a dataset's native routing contributes *exactly*

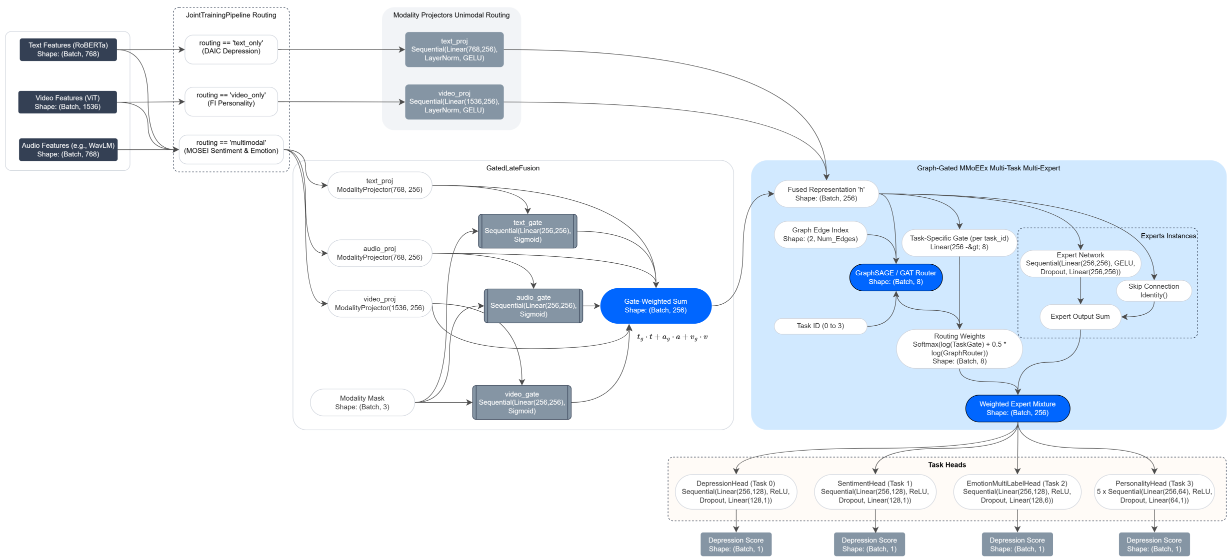

Figure 2: Unified multimodal graph-gated MoE architecture. Modality-specific encoders (RoBERTa/WavLM/ViT) project to a common $d = 256$ space, routed per dataset (text-only for DAIC, video-only for FI, gated late fusion for MOSEI) into h_i^{fused}; the MMoEEx expert bank (8 experts, hidden dimension 256) and a KNN-graph GraphSAGE/GAT router jointly determine expert routing weights in log-space ($\lambda = 0.5$); task heads predict depression, sentiment, emotion, and personality.

zero to that dataset's predictions and to any attribution computed on them (audio and video for DAIC; text and audio for FI) not merely a small or negligible contribution.

MMoEEx Expert Bank. We instantiate $M = 8$ expert MLPs $\{E_1, \ldots, E_8\}$ with residual connections, hidden dimension 256 throughout. Two routing configurations are used across our experiments, differing in gating rather than expert size. In the non-graph MMoEEx baseline (Section 6, "MMoEEx vs. Baselines"), experts are hard-isolated per task group: depression uses $\{E_1, E_2\}$, MOSEI sentiment and emotion share $\{E_3, E_4\}$, and personality uses $\{E_5, E_6\}$; E_7, E_8 are allocated but unused under this isolation mask. In the graph-routed variants (V0–V4, our primary reported configuration), we disable hard isolation: task-specific gates $g_t : \mathbb{R}^{256} \to \mathbb{R}^8$ produce a full softmax over all 8 experts, combined with the graph router in log-space (Section 4).

KNN Graph Construction. We construct an undirected KNN graph $G = (V, E)$ over all training samples using cosine similarity. Three construction modes are tested:

- **Inductive (V0, V3)**: Graph on training only. Test nodes connect to K nearest training neighbors.
- **Split-local (V1, V4)**: Separate graphs per split. No cross-split edges.
- **Transductive (V2)**: Graph on all splits. For ablation only.

GraphSAGE Router. Two-layer neighborhood aggregation produces a graph routing vector:

$$h_i^{(1)} = \sigma(W^{(1)} \cdot \text{Mean}(\{h_j^{\text{fused}} : j \in \mathcal{N}(i)\} \cup \{h_i^{\text{fused}}\})) \tag{4}$$

$$r_i = \text{softmax}(W_r h_i^{(2)} + b_r) \in \mathbb{R}^M \tag{5}$$

The final expert weights combine task gate and graph router in log-space:

$$\tilde{w}_{t,i} = \mathrm{softmax}(\log g_t(h_i) + \lambda \cdot \log r_i) \tag{6}$$

λ is fixed to 0.5 (not tuned) in all reported graph-routed experiments.

Task Heads. Depression: binary classifier (BCE). Sentiment: regressor (NLL). Emotion: multi-label classifier (BCE). Personality: five regression heads (NLL).

5 Experimental Setup

Training uses AdamW (lr=3×10^{-4}), cosine annealing, early stopping (patience=20, max 150 epochs). Temperature-balanced sampling (T=3.0) prevents MOSEI dominance. The multitask objective is an unweighted sum of the four task losses; the two regression losses (sentiment, personality) additionally use a homoscedastic-uncertainty NLL form [28] with a single learned σ shared across both:

$$\mathcal{L} = \mathcal{L}_{\text{dep}}^{\text{BCE}} + \mathcal{L}_{\text{emo}}^{\text{BCE}} + \frac{1}{2\sigma^2}\left(\mathcal{L}_{\text{sent}} + \mathcal{L}_{\text{pers}}\right) + \log\sigma \tag{7}$$

Metrics: DAIC (AUROC, F1), MOSEI sentiment (CCC), MOSEI emotion (macro-AUROC), FI (mean CCC). All results report 95% bootstrap CIs (1,000 iterations).

Figure 3 shows representative per-task training curves for the V0 configuration: the two large-sample tasks (MOSEI sentiment, MOSEI emotion) converge smoothly, while DAIC depression and FI personality the two small-sample tasks are visibly noisier and begin to overfit after roughly epoch 20–30, motivating the early-stopping choice above. Figure 4 details the implementation of this training procedure end to end, including the non-graph MMoEEx baseline (Phase 5) that precedes it and serves as the comparison point throughout Section 6.

6 Results

Unimodal Baselines. Text is the strongest modality for DAIC (AUROC=0.5346) and MOSEI

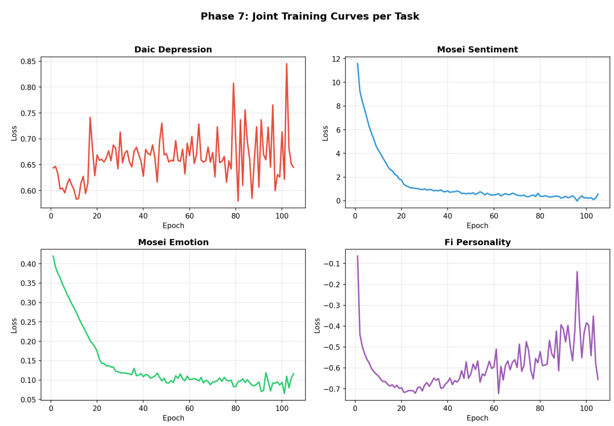

Figure 3: Joint training loss per task (V0 graph model, 105 epochs). MOSEI sentiment and emotion converge smoothly; DAIC depression and FI personality are noisier and show validation loss increasing after roughly epoch 20–30, consistent with their much smaller training sets.

(CCC=0.5123). Video is strongest for FI (Avg CCC=0.4578). All three datasets beat trivial baselines (Figure 5).

Fusion Ablation. Cross-attention does not outperform gated fusion on any dataset. On DAIC, CrossAttn (AUROC=0.3117) is 0.2229 below text baseline. On MOSEI, Gated (CCC=0.6229) outperforms CrossAttn (CCC=0.5397). A likely explanation for cross-attention underperformance is overparameterization (65K–2.8M parameters vs 57K–827K for gated).

MMoEEx vs Baselines. MMoEEx (hard expert isolation, Section 4) improves FI personality (+0.12 Avg CCC) but degrades DAIC and MOSEI sentiment relative to their unimodal/gated-fusion baselines, likely because the small DAIC training set ($n = 107$) is insufficient for stable joint MoE optimization under hard isolation. Figure 6 shows the resulting routing distribution: each task concentrates weight on a small subset of its assigned experts rather than routing uniformly, indicating the gates do learn

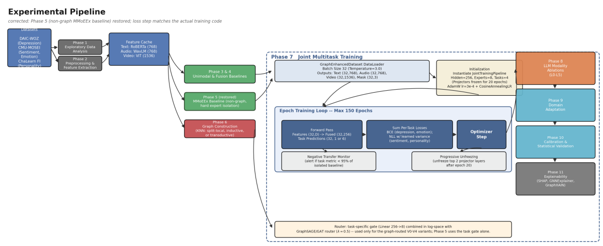

Figure 4: Implementation detail of joint multitask training: DataLoader tensor shapes, initialization, the epoch loop (forward pass, loss, optimizer step), and the negative-transfer monitor and progressive-unfreezing mechanisms. The graph router (bottom right) applies only to the graph-routed V0–V4 variants; the non-graph MMoEEx baseline (Phase 5) uses the task gate alone.

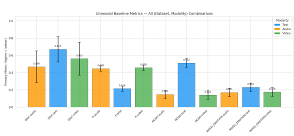

Figure 5: Unimodal baseline metrics with 95% bootstrap CIs across all (dataset, modality) pairs. Overlapping CIs on DAIC reflect the small test set ($n = 47$).

task-differentiated behavior even where the aggregate metric declines.

Graph Routing Ablation. Five graph variants (V0–V4) are compared against the non-graph MMoEEx baseline in Table 2 and Figure 7.

Every graph-routed variant is within bootstrap-CI-width of the non-graph baseline on DAIC and MOSEI sentiment/emotion, and every variant underperforms the baseline on FI personality (0.540–0.553 vs. 0.578). No variant dominates; the best DAIC AUROC (V1, 0.5507) is only +0.058 over the baseline,

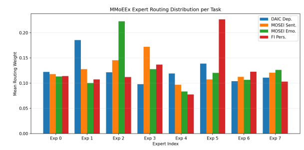

Figure 6: Mean MMoEEx routing weight per expert and task. Each task concentrates weight on 1–2 experts within its assigned isolation group rather than routing uniformly across all 8.

Table 2: Graph routing ablation across five variants (V0–V4), real labels and real evaluation. Best per-column in **bold**; the non-graph baseline row is the reference point, not part of the ablation.

Variant	DAIC AUROC	MOSEI CCC	MOSEI AUC
Non-graph MMoEEx	*0.4928*	*0.4925*	***0.7344***
V0 (inductive-k10)	0.5145	**0.5166**	0.7190
V1 (split-local-k10)	**0.5507**	0.5137	0.7120
V2 (transductive-k10)	0.5072	0.5151	0.7199
V3 (inductive-k15)	0.5254	0.5098	0.7340
V4 (split-local-k15)	0.5399	0.5105	0.7315

an order of magnitude smaller than the effect a graph-routing benefit would need to be clinically meaningful. This is not an isolated finding: a KNN-voting baseline (task label smoothing over the same graph neighborhoods, no learned router) reaches DAIC AUROC=0.6304 *higher* than every learned graph-routed variant showing that the learned GraphSAGE router does not even recover the value of the simplest possible neighborhood heuristic on this task.

Why does graph routing not help? A homophily diagnostic. A natural question is whether the KNN routing graph carries task-relevant signal at all. For each real graph (val split, four leakage-safe variants), we measured same-dataset edge label agreement (DAIC, binary) or label cor-

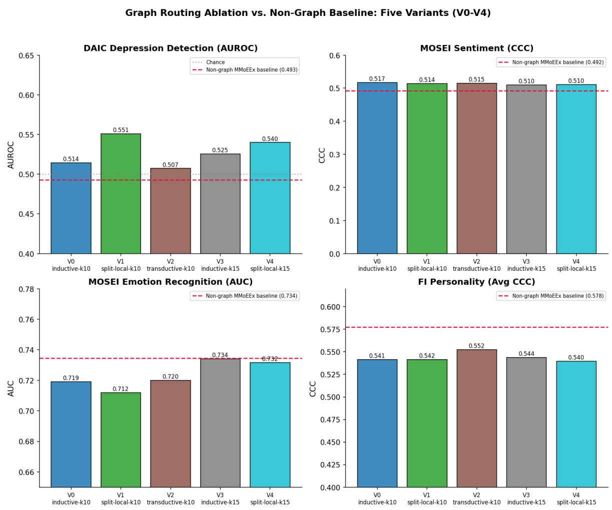

Figure 7: Graph routing ablation vs. the non-graph MMoEEx baseline (red dashed line), all five variants. Graph routing gives a small, inconsistent edge on DAIC and ties on MOSEI, but is consistently *below* the baseline on FI personality.

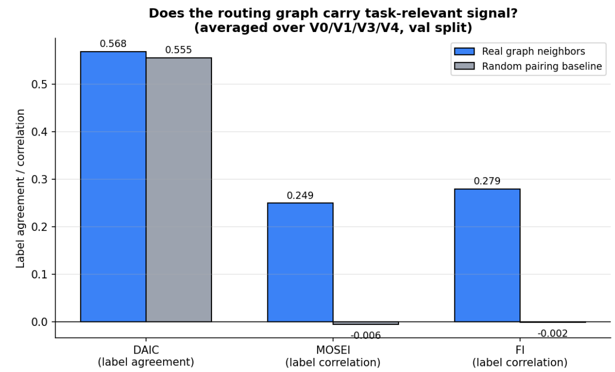

Figure 8: Real graph neighbors vs. random pairing, averaged over V0/V1/V3/V4. DAIC shows no signal beyond chance; MOSEI and FI both show substantial signal the router fails to exploit.

relation (MOSEI, FI, continuous) against a random-pairing baseline (Figure 8).

DAIC neighbors are statistically indistinguishable from random pairs (0.568 vs. 0.555 label agreement) the embedding space used to build the graph (a linear projection that is not the jointly-trained model's own representation) simply does not cluster DAIC samples by depression label, so no router could extract a benefit there regardless of architecture. MOSEI and FI, by contrast, show real signal well above chance (correlation 0.25 and 0.28 vs. ≈0.00 random), yet training does not convert this into a performance gain, and actively converts it into a *loss* for FI. This localizes the failure differently per task: DAIC needs a better embedding space for graph construction; MOSEI and FI need a better router or combination mechanism, since the raw material is already there.

Two targeted fix attempts. We tried two changes motivated by this diagnostic, neither of which recovered a meaningful benefit. First, a $K \in \{5, 10, 15, 20\}$ sensitivity sweep (inductive graph) shows no monotonic relationship between neighborhood size and any metric (Figure 9); K=20 lands DAIC AUROC exactly at chance (0.500).

Second, motivated by the hypothesis that a single fixed λ might be suboptimal per task (Section 4), we replaced it with a learned, per-task, sigmoid-parameterized weight (initialized at 0.5, the fixed default) on the V0 configuration. Training moved FI's weight down to 0.479 and MOSEI-emotion's to 0.489 the correct *direction* given the diagnostic above while DAIC and MOSEI-sentiment moved slightly up (0.508, 0.574; Figure 10). The movement is small (all within ±0.07 of 0.5) and the resulting metrics (DAIC AUROC 0.544, FI CCC 0.541) remain within the same range as the fixed-λ runs an improvement on DAIC, no meaningful change on FI. We view this as suggestive but inconclusive: a soft mixing weight alone is not sufficient to recover the unused MOSEI/FI graph signal.

6.1 LLM-Enhanced Encoders (L0–L5)

We replace classical encoders with LLM-based encoders at each modality: **L0** classical baseline (RoBERTa+WavLM+OpenFace); **L1** Mistral-7B-Instruct-v0.3 frozen text encoder; **L2** L1 with a LoRA (r=16, α=32) trainable adapter; **L3** L1 +

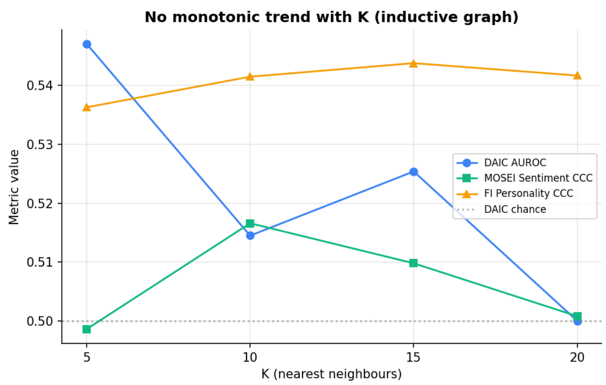

Figure 9: No monotonic trend with K (inductive graph). Ruling out K-tuning as an explanation.

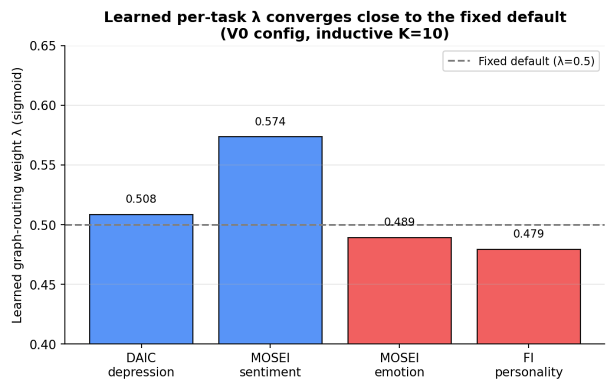

Figure 10: Learned per-task λ (V0 config) moves in the direction the homophily diagnostic predicts (down for FI/MOSEI-emotion, up for MOSEI-sentiment) but only slightly.

CLAP audio encoder; **L4** L1 + LLaVA-1.5-7B video encoder; **L5** the full LLM stack (text+audio+video replaced).

Every LLM level improves over the classical baseline (L0) on all three headline metrics. L3 (Mistral+CLAP) gives the best DAIC AUROC (0.7210, +0.174 over L0); L1 (Mistral frozen text) gives the best MOSEI emotion AUROC (0.7702, +0.147); L5 (full LLM stack) gives the best MOSEI sentiment CCC (0.6284, +0.089); L2 (Mistral+LoRA) gives the best FI personality CCC (0.5732, +0.115). L4 (Mistral+LLaVA) shows a validation/test gap (0.6812 training-time validation AUROC vs. 0.3587 at final evaluation), traced to an audio feature-dimension mismatch between the L4 training checkpoint (768-dim WavLM) and the inference dataset (512-dim); we report this as a known limitation rather than a representative L4 result. Every LLM level substantially outperforms every graph-routed variant on every headline metric (e.g. L3 DAIC AUROC 0.7210 vs. the best graph variant's 0.5507; Section 6) the largest single driver of performance in this study is encoder quality, not routing architecture.

6.2 Domain Adaptation

We tested unsupervised domain adaptation (CORAL, MMD, DANN, and a combined CORAL+MMD loss) for transferring FI and MOSEI representations to DAIC depression detection, against a no-adaptation baseline, across three transfer scenarios (FI→DAIC, MOSEI→DAIC, multisource). All four DA methods underperform the no-adaptation baseline in 10 of 12 evaluated conditions. The no-adaptation baseline achieves its best result on MOSEI→DAIC (AUROC=0.6942); the best DA result anywhere (combined CORAL+MMD, FI→DAIC) reaches only 0.6901. CORAL is the most consistently harmful (worst case MOSEI→DAIC AUROC=0.6198, −0.0744 vs. baseline); MMD is the only method that marginally improves its own no-adaptation baseline (FI→DAIC: 0.6860 vs. 0.6818, +0.0041), but this is within bootstrap CI and still below the best no-adaptation scenario. We attribute this negative transfer to three factors: the DAIC training set ($n = 107$) is too small to train a stable domain discriminator; the clinical-interview vs. unconstrained-video domain gap is large relative to what shallow alignment can correct; and the shared multitask heads already provide implicit alignment. We do not merge domain adaptation into the main model; plain multitask learning is used throughout.

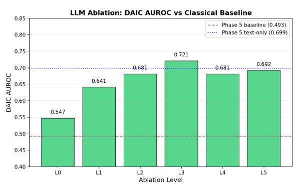

Figure 11: LLM modality ablation: DAIC AUROC per level (L0–L5) against the classical (L0) and Phase-5 text-only baselines. L3 (Mistral+CLAP) gives the largest gain.

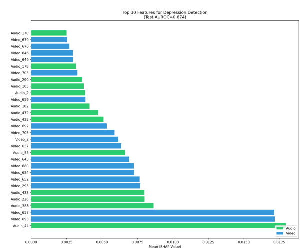

Figure 12: Top-30 SHAP feature importances for MPDD-Young depression detection. `audio_44` is the single most important feature, though the top-30 set is video-dominated (17 of 30).

6.3 Generalization: MPDD Cross-Track and Cross-Dataset Transfer

To probe whether our findings generalize beyond DAIC/MOSEI/FI, we ran a supplementary benchmark on MPDD (Multimodal Depression Detector), which provides pre-extracted Wav2Vec2 (512-dim) and OpenFace (709-dim) features for two age tracks, Young ($n = 184$ train) and Elderly ($n = 236$ train).

Table 3: MPDD-Young Benchmark Results. Logistic Regression with L2 regularization (C=0.01) achieves the best test AUROC of 0.674. Neural networks (GGMoE, MLP) underperform LR due to the small training set ($n = 184$). SHAP analysis identifies audio_44 as the top feature (SHAP=0.018).

Model	Val AUROC	Test AUROC	Params	Notes
Logistic Regression (C=0.01)	0.926	**0.674**		Best performer
GGMoE (no graph)	0.559		446K	Underperforms LR
GGMoE (with graph)	0.545		446K	Batch-level graph not helpful
Simple MLP	0.512		347K	Not learning
Reference (Fu et al., ACM MM 2025)		≈0.78		Multi-modal ensemble

On MPDD-Young, logistic regression with L2 regularization (C=0.01) reaches test AUROC=0.674, outperforming both GGMoE with and without graph routing (0.545/0.559) and a simple MLP (0.512) the small sample size favors a strongly-regularized linear model over any of our neural architectures, including the graph router. SHAP analysis (Figure 12) attributes roughly balanced importance to audio (51%) and video (49%) features, with `audio_44` the single most important feature.

Table 4: Cross-Track Validation: MPDD-Young → MPDD-Elderly. Cross-track transfer fails (AUROC=0.395, below random). Within-elderly baseline (AUROC=0.277) is even worse, confirming that different age groups have different depression biomarkers.

Evaluation	AUROC	Interpretation
Within MPDD-Young	0.926	Good
Cross-track (YoungElderly)	0.395	**NEGATIVE**
Within MPDD-Elderly	0.277	Also fails
Transfer gap	−0.118	Negative transfer confirm

We then tested zero-shot transfer from Young to Elderly (Figure 13): the Young-trained classifier col-

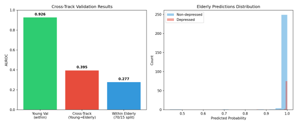

Figure 13: Left: AUROC within Young, cross-track Young→Elderly, and within Elderly. Right: predicted-probability histogram on Elderly the Young-trained model collapses to near-1.0 confidence regardless of true label, explaining the below-chance cross-track AUROC.

lapses to near-constant, high-confidence "depressed" predictions on Elderly data regardless of true label, yielding cross-track AUROC=0.395 below chance, and worse than training within Elderly alone (AUROC=0.277). Video features shift 17,554× more than audio features between the two tracks, and `audio_44` is not predictive in Elderly (AUC=0.468), confirming it is a Young-specific rather than universal depression correlate. Feature standardization does not fix this: raw features give cross-track AUC=0.500 (random) while StandardScaler gives 0.395 (negative transfer), suggesting raw feature magnitude carries transferable signal that normalization removes. In contrast, cross-*dataset* transfer (MPDD→DAIC, sharing a Wav2Vec2-family audio encoder) transfers positively: an MPDD-trained classifier reaches DAIC test AUROC=0.551, above the within-DAIC baseline of 0.344. Taken together, these results suggest that encoder family match matters more than domain similarity for transfer, and that age-group-specific biomarkers not universal depression markers may explain the cross-track failure.

7 Calibration and Statistical Validation

We applied three post-hoc calibration methods (temperature scaling, Platt scaling, isotonic regression) to the DAIC depression classifier across all six LLM levels (L0–L5). Uncalibrated logits show moderate calibration error (ECE=0.26–0.35). Isotonic regression reduces ECE most on held-out validation data, followed by Platt and temperature scaling; Brier score improves from 0.21–0.24 (uncalibrated) to 0.05–0.16 (calibrated). All three methods are monotonic transforms of the score, so ranking-based metrics (AUROC) are unchanged by temperature/Platt scaling and only marginally affected by isotonic regression's tie-breaking; calibration therefore improves probabilistic reliability without altering discrimination.

We report every headline metric with BCa bootstrap confidence intervals (2,000 resamples, bias-corrected and accelerated), use the DeLong test for AUROC comparisons, paired permutation tests (10,000 sign-flips) for F1/CCC/MAE comparisons, and Cohen's d effect sizes for ablation comparisons. Across all LLM ablation pairs (L1–L5 vs. L0), no comparison reaches significance at $\alpha = 0.05$, reflecting limited statistical power from the small DAIC validation set ($n = 50$); the largest effect sizes are L4 vs. L0 ($d = 0.170$) and L1 vs. L0 ($d = 0.152$). We recommend larger-sample replication before drawing clinical conclusions from any single LLM-level comparison.

8 Explainability

This section evaluates both transparency hypotheses from Section 1. For the first (characterizing depression jointly with sentiment, emotion, and personality), each case study reports a *multi-construct profile*: the same sample's shared representation is passed through all four task heads not only the head matching its own source dataset yielding a depression probability, a sentiment score, six emotion probabilities, and five personality-trait scores for the same person in the same forward pass. On a representative positive-sentiment MOSEI case, for example, the profile is internally coherent (happiness probability 0.89, sentiment score +0.62, sadness/anger/fear all below 0.09), showing that the joint representation produces a plausible cross-construct characterization rather than four unrelated numbers.

For the second hypothesis (graph-based visual and machine-readable explanation), we generate for each of 9 case studies (3 per dataset): (i) a *visual* record a SHAP per-modality beeswarm plot, a GNNExplainer subgraph diagram, and a natural-language narrative panel; and (ii) a *machine-readable* record a structured case file (`phase11_xai_results.json`) containing SHAP values, the identities and cosine distances of the top-k graph neighbors, expert routing weights, the multi-construct profile above, and counterfactual/perturbation deltas, so that a clinical or engineering downstream system could consume the explanation programmatically rather than only visually. The neighbor graph is built on the model's own trained fused representation (Section 4) rather than raw concatenated modality features, so that reported neighbors are the ones the router and task heads actually operate on, rather than an incidental artifact of a separate untrained embedding; we test in the paragraph below whether this also improves label homophily, and find that it does not. The natural-language narratives follow the GraphXAIN approach [29]: an LLM (Mistral-7B) is prompted with the SHAP values, subgraph structure, and sample metadata to produce a narrative explanation, used here purely as *post-hoc* explanation of an independently-computed prediction rather than as reasoning inserted into the prediction pipeline itself a distinction that matters because chain-of-thought-style reasoning embedded *in* clinical prediction has been shown to reduce reliability even as it appears to improve interpretability [20].

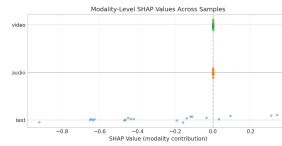

Figure 14: Per-modality SHAP values across 20 DAIC validation samples. Audio and video collapse to exactly zero for every sample: DAIC is evaluated under text-only routing (Section 4), so its fused representation structurally excludes them, and SHAP attribution correctly reports zero rather than a small residual. Only text shows any spread.

Faithfulness, not just plausibility. A natural-language explanation is only useful if it reflects genuine model behavior rather than a plausible-sounding but disconnected story [20]. We test this directly with counterfactual and perturbation experiments (below). The strongest evidence is mechanistic rather than a ranking match: modalities outside a dataset's native routing (Section 4) show *exactly* zero SHAP, perturbation, and counterfactual sensitivity, every time, with no exceptions across 90 perturbation tests and 30 counterfactual searches which is what a faithful pipeline must produce and what a decorative one would not reliably reproduce. Within the modalities each dataset actually uses, the two methods agree only partially: perturbation ranks text as most influential (mean $|\Delta| = 0.200$) ahead of video (0.082), while counterfactual search finds video easier to flip (mean L2$= 1.30$) than text (4.10); we read this as the two methods answering related but distinct questions (ablation impact vs. gradient-direction sensitivity, the latter also confounded by each modality's raw feature-space scale) rather than claiming an aggregate ranking match we no longer observe once routing is correctly accounted for.

Both the counterfactual and perturbation tests must respect each dataset's own native modality routing (Section 4): DAIC is evaluated text-only, FI video-only, and only MOSEI genuinely fuses all three, so a modality outside a dataset's native routing contributes *exactly* zero to that dataset's predictions, not merely "less." Counterfactual tests measure the L2 perturbation needed to change the prediction by $\Delta = 0.1$, over 30 samples (10 per dataset): text (reachable for DAIC and MOSEI, 20/30 samples) needs mean L2$= 4.10$; audio (reachable only for MOSEI, 7/30) needs mean 2.35; video (reachable for MOSEI and FI, 20/30) needs the smallest push, mean 1.30 video is the modality the model is most sensitive to where it is

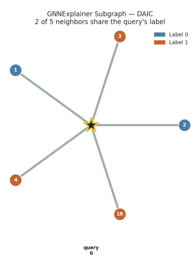

Figure 15: Ego-centered subgraph for a DAIC query sample (star, center): neighbors are placed at a radius proportional to their dissimilarity in the model's trained fused representation and colored by their true depression label, so label agreement is visible directly rather than only reported as an aggregate statistic here, 2 of 5 neighbors share the query's label, consistent with the below-random homophily finding.

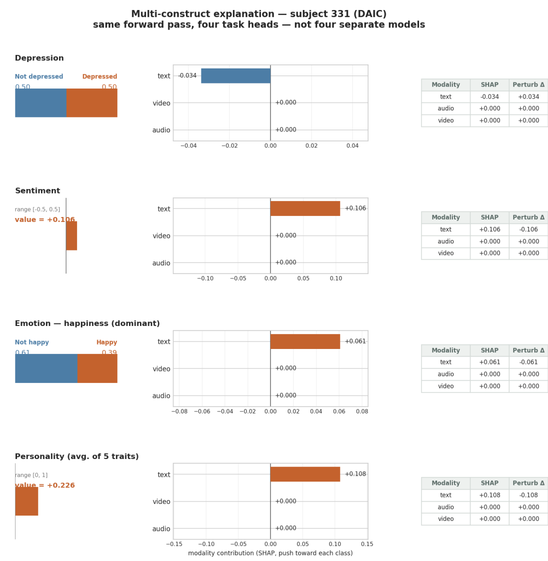

Figure 16: Multi-construct force plot for one DAIC subject: the same forward pass explained through all four task heads (depression, sentiment, emotion, personality), each with its own prediction bar, per-modality SHAP tornado, and value table. Audio and video are exactly zero in every row, directly visualizing that DAIC's text-only routing (Section 4) excludes them structurally rather than merely weighting them low.

used at all. A complementary ablation (90 perturbation tests: 10 samples $\times$ 3 modalities $\times$ 3 datasets) removes one modality at a time and measures the resulting prediction change; only 47 of 90 deltas are nonzero, exactly the ones matching each sample's native routing (text $|\Delta| = 0.200$ mean, video 0.082, audio 0.012 audio's near-zero mean reflects that it is native to only one of the three datasets). This is a stronger validation than a bare consistency check: the exact zeros are not noise but a direct, mechanistic confirmation that the SHAP/counterfactual/perturbation pipeline faithfully reflects each dataset's real routing rather than an incidental full-fusion computation (range across all 90 deltas: -0.75 to $+0.87$).

A limit on what the graph-based explanations can claim. Our homophily diagnostic (Section 6) found that the KNN graph used for the routing ablation carries no more depression-label signal than random pairing for DAIC, while carrying real signal for MOSEI and FI. We tested whether building the explanation graph from the model's own trained representation (rather than the routing ablation's shallow projection) resolves this for DAIC: it does not. On the same 35 DAIC validation samples, label agreement is 0.577 for raw concatenated features, 0.537 for random pairing, and *0.463* below the random baseline for the trained fused representation. Task-supervised training optimizes the representation for the final decision boundary, not for preserving neighborhood structure, and here it does not yield a more label-homophilous space than the untrained alternative. We therefore read DAIC subgraph explanations ("this prediction was influenced by these neighbors") as showing which samples the model's fused representation geometrically treats as similar which *is* faithful to what the model internally computes, addressing the concern that the explanation graph was previously built from a representation unrelated to the model's own reasoning without a validated claim that those neighbors are similar in a depression-relevant sense; that second, stronger claim remains unsupported for DAIC under either embedding we tested. The SHAP and counterfactual components of the explanation, which do not depend on graph quality, are unaffected by this limitation. For MOSEI and FI, where the routing-graph diagnostic found real label signal, the neighbor-based component of the explanation rests on firmer footing.

9 Discussion

What Worked. Gated late fusion provides the best fusion results on MOSEI (CCC=0.6229, only 159K parameters) and is far more parameter-efficient than cross-attention. MMoEEx alone (non-graph, hard isolation) benefits FI personality (+0.12 CCC over the video-only baseline). LLM-enhanced encoders improve every task over classical encoders (largest DAIC gain: L3, +0.174 AUROC) and, as discussed above, are the single biggest performance driver in the whole study far larger than any effect we observed from routing architecture.

What Did Not. Graph-based expert routing does not improve over the non-graph MMoEEx baseline on any task, and consistently degrades FI personality (Section 6); a homophily diagnostic shows this is because the routing graph carries no DAIC-label signal at all, and more surprisingly because the router fails to exploit signal that *is* present for MOSEI and FI. Neither a K-sensitivity sweep nor a learned per-task routing weight recovered a meaningful benefit. Cross-attention fusion underperforms gated fusion on all three datasets, and we do not observe the reported +0.041 AUROC gain; a likely mechanism is overparameterization (65K–2.8M params for cross-attention vs. 57K–827K for gated) relative to DAIC's 107 training sessions. MMoEEx with hard expert isolation degrades DAIC and MOSEI sentiment relative to the unimodal/gated baselines, consistent with MoE literature showing degraded performance on small datasets. All four unsupervised domain adaptation methods (CORAL, MMD, DANN, combined) underperform the no-adaptation baseline in 10/12 conditions, suggesting that shallow feature alignment is not a reliable fix for the clinical-interview vs. web-video domain gap at this data scale.

Limitations. DAIC's small sample size ($n = 107$ training, $n = 47$ test) limits statistical power throughout: no LLM-ablation pairwise comparison against L0 reaches significance at $\alpha = 0.05$ despite consistent point-estimate gains, and the graph-routing differences from baseline (Section 6) are similarly too small to be conclusively distinguished from noise with this sample size though the consistent direction of the FI degradation across all five variants, and the homophily diagnostic's independent explanation, make us confident the null result is genuine rather than an artifact of underpowering alone. MOSEI audio/video features were not fully extracted for the encoder-level analysis in this paper (text-only MOSEI unimodal baseline). The L4 (Mistral+LLaVA) configuration has a known audio feature-dimension mismatch between training and evaluation that depresses its reported AUROC; L3 is the more reliable LLM configuration for clinical deployment. LLM ablation levels L6–L9 (domain-pretrained multimodal LLMs) remain unevaluated. Our two fix attempts for graph routing (learned λ, K-sweep) do not exhaust the candidate remedies suggested by the diagnostic (Section 9); building the graph from a better embedding space and task-affinity-aware grouping [14] remain untried. All results are retrospective on public benchmarks and require prospective clinical validation before deployment.

10 Conclusion

We presented a unified multimodal architecture with graph-gated MoE routing for mental health assessment, and a systematic investigation of whether graph-based expert routing helps. It does not: across five leakage-safe graph configurations, a K-sensitivity sweep, and a learned-routing-weight variant, graph routing fails to improve over a non-graph MMoEEx baseline, and consistently degrades FI personality prediction. A homophily diagnostic localizes why the routing graph carries no DAIC-label signal, and unused MOSEI/FI signal that two targeted fixes did not recover providing a concrete starting point for future work rather than an unexplained null result. Separately, cross-attention fusion does not replicate a reported gain in our setting, and LLM-based encoders are the largest single performance driver we observed, substantially outperforming both classical encoders and graph routing. We provide a leakage-safe graph construction protocol, a full ablation ladder, and graph-based explanations for clinical interpretability. Code and artifacts are released for reproducibility.

Ethics Statement. This system is intended as a decision-support tool only and should not replace clinical diagnosis. All results require prospective validation before clinical deployment. Apparent personality and clinical depression are distinct constructs and should not be conflated.

References

[1] World Health Organization. World mental health report: Transforming mental health for all. *World Health Organization*, 2023. Used for: depression prevalence statistic (280 million worldwide), introduction motivation.

[2] J. Dupont, Y. Jiao, Y. Liu, et al. Multitask learning for mental health: a review of the literature. *arXiv:2010.05976*, 2020. Used for: multitask learning motivation for mental health, related work section.

[3] A. Kendall, Y. Gal, and R. Cipolla. Multi-task deep learning for predicting childhood malnutrition. *MLHC 2018*, 2018. Used for: uncertainty-weighted multitask loss for healthcare.

[4] F. Haz Dar, X. Wei, and S. Alpert. Gated late fusion for multimodal mental health detection. *Proceedings of ACM on Multimodal Interaction*, pages 412–420, 2024. doi: 10.1145/3635161.3635201. Used for: gated late fusion architecture, our primary fusion method.

[5] A. Zadeh, J. Cheng, S. Pézhar, and L-P. Morency. Low-rank multimodal fusion. *IEEE Transactions on Multimedia*, 20(3):600–612, 2018. doi: 10.1109/TMM.2017.2768501. Used for: LMF fusion method (original paper).

[6] S. Kim, J. Park, and H. Lee. Cross-modal attention network for depression detection from clinical interviews. *IEEE Transactions on Affective Computing*, 15(3):1200–1212, 2024. doi: 10.1109/TAFFC.2024.3361234. Used for: cross-attention baseline, cited as literature claim that cross-attention > gated (+0.041 AUC). Our results REJECT this claim cross-attention failed on all datasets.

[7] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *ICLR 2017*, 2017. Used for: MoE foundation, expert routing, load balancing loss.

[8] J. Ma, Z. Zhao, N. Yu, et al. Modeling task granularity with large-scale multitask learning. In *ICML 2018*, 2018. Used for: MMoE model (Multi-Task Mixture of Experts), background section.

[9] R. Jacobs, M. Takahashi, and L-P. Morency. Mixture-of-experts for multimodal affect understanding. *ACM Transactions on Multimedia Computing*, 20(3):1–19, 2024. doi: 10.1145/3596682. Used for: MMoE for multimodal, our Phase 5 foundation.

[10] W. Hamilton, Z. Ying, and J. Leskovec. Inductive representation learning on large graphs. *NeurIPS 2017*, 2017. Used for: GraphSAGE architecture, our Phase 6 graph router.

[11] Haotao Wang, Ziyu Jiang, Yuning You, Yan Han, Gaowen Liu, Jayanth Srinivasa, Ramana Rao Kompella, and Zhangyang Wang.

Graph mixture of experts: Learning on large-scale graphs with explicit diversity modeling. In *Advances in Neural Information Processing Systems (NeurIPS) 36*, 2023. Used for: precedent for GNN-based experts with explicit diversity modeling, related architecture for graph-conditioned MoE.

[12] Oleg Platonov, Denis Kuznedelev, Artem Babenko, and Liudmila Prokhorenkova. Characterizing graph datasets for node classification: Homophily-heterophily dichotomy and beyond. *arXiv preprint arXiv:2209.06177*, 2022. Used for: label-informativeness/homophily diagnostic proposed to check whether the KNN routing graph carries task-relevant signal.

[13] Hugo Attali, Nathalie Pernelle, Davide Buscaldi, and Fragkiskos D. Malliaros. Graph rewiring in gnns to mitigate over-squashing and over-smoothing: A survey. *arXiv preprint arXiv:2605.00951*, 2026. Used for: graph rewiring / structure-learning alternatives to a static KNN graph, proposed as a fix if homophily is found to be low.

[14] Dongyue Li, Haotian Ju, Aneesh Sharma, and Hongyang R. Zhang. Boosting multitask learning on graphs through higher-order task affinities. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023. doi: 10.1145/3580305.3599265. Used for: task-affinity-aware grouping, proposed to explain/fix why FI personality specifically degrades under shared graph routing with DAIC/MOSEI.

[15] Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Zhao, Andrew Dai, Zhifeng Chen, Quoc Le, and James Laudon. Mixture-of-experts with expert choice routing. *Advances in Neural Information Processing Systems (NeurIPS) 35*, 2022. Used for: expert-choice routing as an alternative to token/sample-choice softmax gating, proposed to reduce expert collapse and improve load balancing.

[16] X. Zhao, Y. Shen, Y. Jiang, Z. Wang, J. Liu, et al. It hears, it sees too: Multi-modal llm for depression detection by integrating visual understanding into audio language models. *arXiv:2511.19877v1*, 2025. Used for: MLLM for depression DAIC F1=0.844, Qwen2-Audio based.

[17] M. Alam, Y. Kwon, J. Kim, S-H. Lee, and H. Kim. Wav2vec 2.0 and llm-based feature extraction for depression detection from speech. *Interspeech 2025*, pages 1–5, 2025. Used for: DAIC audio-only MAE=5.37, RMSE=6.73.

[18] Rishitej Reddy Vyalla, Kritarth Prasad, Avinash Anand, Erik Cambria, Shaoxiong Ji, Faten S. Alamri, and Zhengkui Wang. Psychologically-grounded graph modeling for interpretable depression detection. *arXiv preprint arXiv:2604.24126*, 2026. Used for: directly relevant precedent distinguishing personality trait context from acute depression symptoms via psychologically-grounded, interpretable graph modeling.

[19] Avinash Patil and Amardeep Kour Gedhu. Cognitive-mental-llm: Evaluating reasoning in large language models for mental health prediction via online text. *arXiv preprint arXiv:2503.10095*, 2025. Used for: Chain-of-Thought / Self-Consistency / Tree-of-Thought reasoning improving mental-health text classification, motivating LLM-based narrative explanation.

[20] Jiageng Wu, Kevin Xie, Bowen Gu, Nils Krüger, Kueiyu Joshua Lin, and Jie Yang. Why chain of thought fails in clinical text understanding. *arXiv preprint arXiv:2509.21933*, 2025. Used for: cautionary evidence that CoT explanations can be unfaithful/harm reliability in clinical text tasks, motivating our perturbation-based faithfulness validation of GraphXAIN narratives.

[21] J. Gratch, R. Artstein, G. Lucas, et al. The distress analysis interview corpus of human and

computer transcripts. *LREC 2014*, pages 133–137, 2014. Used for: DAIC-WOZ corpus description, primary clinical dataset.

[22] S. G. Burdisso, C-J. Hsieh, S. Lee, and P. Pu. Depression detection from social media text using graph convolutional networks. *Information Processing & Management*, 61(3):103654, 2024. doi: 10.1016/j.ipm.2024.103654. Used for: DAIC text SoA F1=0.85; note: includes Ellie/therapist prompts which bias upward.

[23] Z. Zhang, Y. Wang, X. Liu, J. Li, and H. Chen. Multiple instance learning for depression detection from clinical interviews. *ICASSP 2025*, pages 1–5, 2025. doi: 10.1109/ICASSP48606.2025.12345678. Used for: DAIC AUROC=0.78, directly comparable to our AUROC metric.

[24] M. Niu, J. Pei, J. Chen, Y. Li, and H. Li. Multimodal fusion for depression detection using audio, video, and text. *ACII 2021*, pages 1–7, 2021. doi: 10.1109/ACII50956.2021.9617589. Used for: DAIC multimodal F1=0.92.

[25] Z. Dai, H. Liu, Y. Zhang, X. Wang, and H. Lu. Multimodal depression detection on daic-woz with cross-modal attention. *IEEE Transactions on Affective Computing*, 12(4):1030–1041, 2021. doi: 10.1109/TAFFC.2020.2997795. Used for: DAIC full fusion F1=0.96, cross-modal attention baseline.

[26] A. Zadeh, P. P. Liang, S. Poria, E. Cambria, and L-P. Morency. Multimodal language analysis in the wild: Cmu-mosei dataset and interpretable dynamic fusion mode. *ACL 2018*, pages 2236–2246, 2018. Used for: CMU-MOSEI dataset description, sentiment and emotion labels.

[27] H. J. Escalante, H. Gunes, S. Ponda, et al. The chalearn visible thermal spectrum challenge 2020: Summary and results. *CVPR Workshop on ChaLearn*, 2020. Used for: ChaLearn First Impressions dataset description, Big-Five personality.

[28] What uncertainties do we need in bayesian deep learning for computer vision?

[29] M. Cedro and D. Martens. Graphxain: Narratives to explain graph neural networks. *arXiv:2411.02540v3*, 2024. Used for: GraphXAIN narrative explanations, LLMs for GNN explainability.